JOHN PENG

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ECE student (B.S. May 2028) building across robotics, embedded firmware, and AI software: ROS 2 control of CAN actuators, nRF52840/ESP32 firmware in C/C++, PCB bring-up, and LLM/RAG pipelines. Seeking Summer 2027 internships.

EDUCATION

University of Southern California | *B.S. Electrical and Computer Engineering, GPA: 3.59* Los Angeles, CA | Expected May 2028

- Courses: Embedded Systems, Data Structures & Algorithms (C++: classes, STL, memory management), Digital Logic & RTL Design, Distributed Systems / IoT, Linear Circuits

SKILLS

Languages: Python, C/C++, Kotlin, Swift

Embedded & Hardware: nRF52840, ESP32, Arduino, Raspberry Pi, Jetson, J-Link, Altium, KiCad, PCBA, bring-up, oscilloscopes, RF sniffer; CAN, BLE, ESP-NOW, UART

Robotics: ROS 2, MoveIt, Gazebo, URDF, MIT-mode torque control, BLDC actuators, OptiTrack, AprilTags, SolidWorks

AI & Software: LangChain agents, RAG / vector DBs, knowledge graphs (KuzuDB, Neo4j), LLM APIs, OpenCV, YOLO, Flask, Docker, Android (AIDL/Binder)

EXPERIENCE

Terra Labs, USC – Tether | *Hardware Project Manager and Co-Project Manager* 09/2026–Present

- Co-lead (with the software PM) a ~8-person mechanical, electrical, and software team turning a 7-DOF arm into an untethered inchworm robot that walks and climbs; built a roadmap of six validation gates (G0–G5).
- Simplified the robot from 5 modules to 3 self-contained wireless modules, each with its own ESP32-class microcontroller and no wiring across modules; chose ESP-NOW first for motor coordination.
- Designed the power architecture (independent 5,000 mAh battery per module, ~2 h target runtime) and dropped the BMS by relying on the motors' low-voltage shutoff plus a per-module DC-DC converter.
- Compared three joint wire-routing strategies in simulation; only the slip-ring design completed the ladder climb and table loop, driving the 360° continuous-rotation middle joint.
- Set up a shared source-of-truth repo with a numbered decision log, risk register, and specs tagged verified or assumed.

SRAM LLC | *Advanced Development Engineering Intern* 06/2026–08/2026

- Developed nRF52840 firmware in C and C++ for a wireless controller replacing a three-state AC interface, with an interlock that prevents both relays from closing and routes every state change through the disconnected state.
- Reverse-engineered the undocumented AC interface, then designed the compact Altium PCB and prepared it for PCBA (Gerbers, BOM, pick-and-place); the board was manufactured, assembled, and tested successfully.
- Root-caused a silent Android AIDL/Binder transaction-ID mismatch between a client library and the Karoo service by diffing generated dispatch tables, leading to a fixed library release; built Kotlin Monitor Message tooling for firmware engineers.
- Ported a Flutter/Android computer-vision app to iOS (Swift, Objective-C++) in under a day using an agentic coding tool.

Manycore Tech | *Machine Learning Engineer Intern (LLM/RAG)* 05/2025–07/2025

- Built a knowledge-graph-augmented RAG pipeline (Alibaba Bailian, KuzuDB) with LangChain multi-hop agents, improving retrieval relevance by 40% and cutting latency by 20%.

Terra Labs, USC – Cove (7-DOF Robotic Arm) | *Software and Electrical Engineer* 02/2026–05/2026

- Built ROS 2 control over CAN for 7 Damiao BLDC actuators: MIT-mode torque control and a hardware/software e-stop with heartbeat timeout.

USC RoboLAND – Robot Locomotion and Navigation Dynamics Lab | *Undergraduate Researcher* 08/2024–Present

- Redesigned multi-legged gait phase control with normalized offsets; ran 46 OptiTrack-tracked trials with Python analysis workflows.

PROJECTS

Circuit Sensei – AI Hardware Agent | *LA Hacks 2026* 04/2026

- Built an AI agent (Gemma, Gemini Vision) that turns a natural-language circuit goal into a webcam-verified Arduino circuit, then generates code, uploads it over serial, and runs electrical tests.

Vyz – Sensory Adaptation XR Headset | *Hardware Project Winner, LA TechWeek AI/ML Buildathon* 10/2025

- Built a Jetson-powered headset with multithreaded Python (OpenCV, SciPy, Flask) and a GPT-3.5 intervention engine reacting in under 200 ms.

USC Makers – SkateMo Autonomous Skateboard | *Technical Project Lead* 08/2025–05/2026

- Led a 6-member team building a self-driving skateboard; built OpenCV/YOLO obstacle avoidance with camera/IMU fusion.

LEADERSHIP & AWARDS

USC Makers, Director of Technical Resources (05/2026–Present): Lead technical resource development (tutorials, documentation, knowledge systems) for a 100+ member engineering club.

Awards: Regeneron ISEF 2023: First Award in Embedded Systems, EU Contest for Young Scientists Award, SAE International Best Projects Special Award; Conrad Challenge Innovation Summit Finalist (2023)